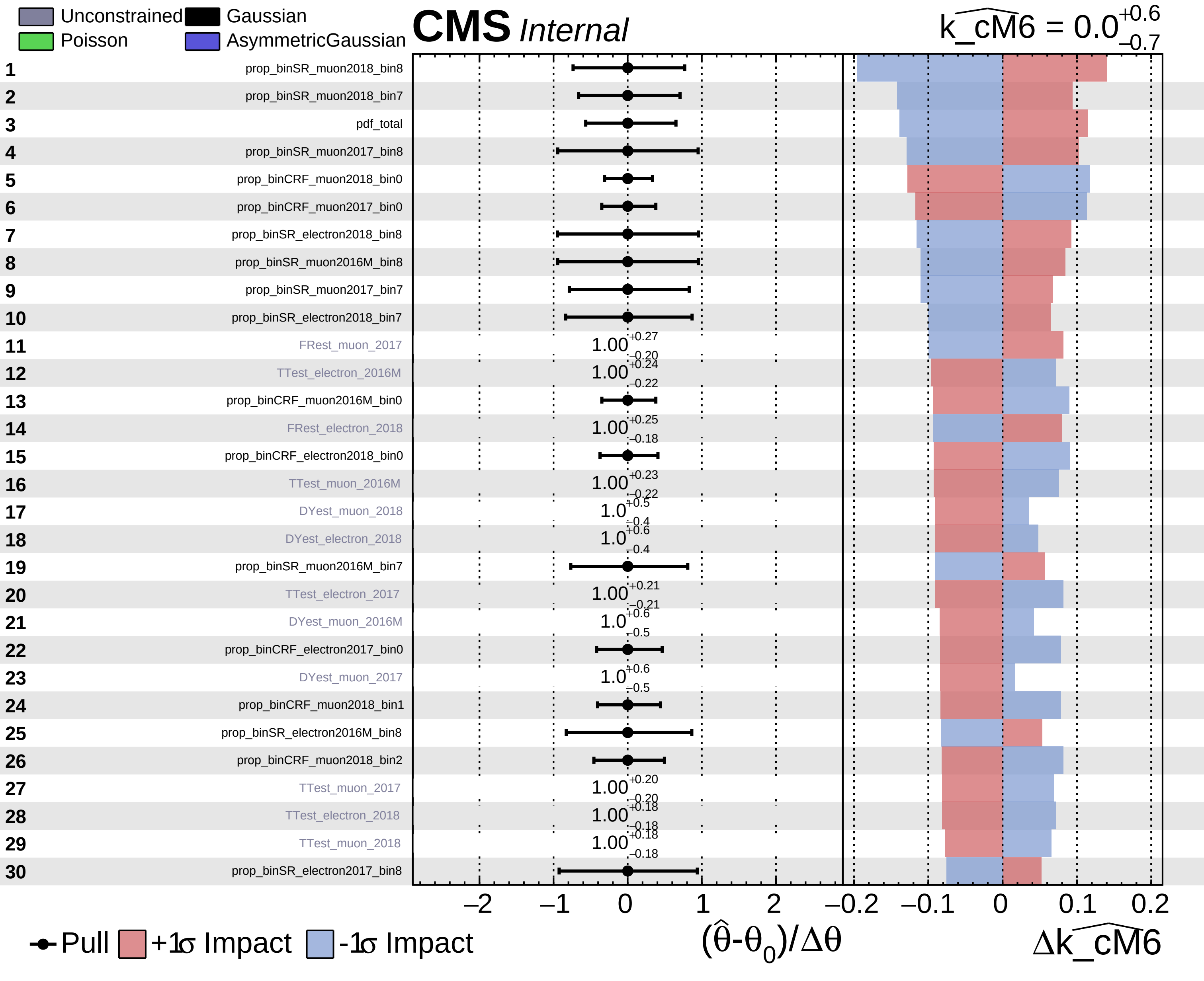
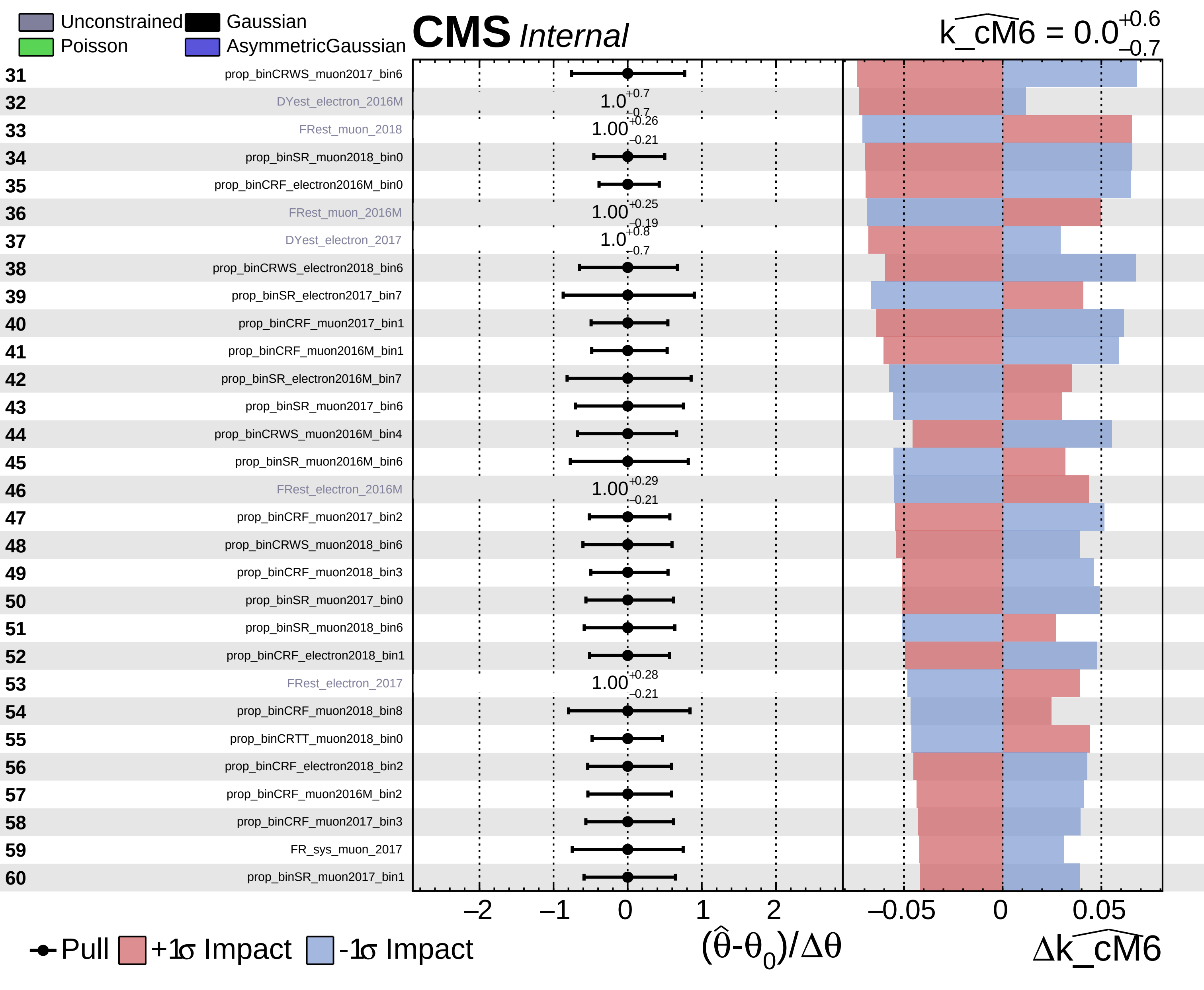
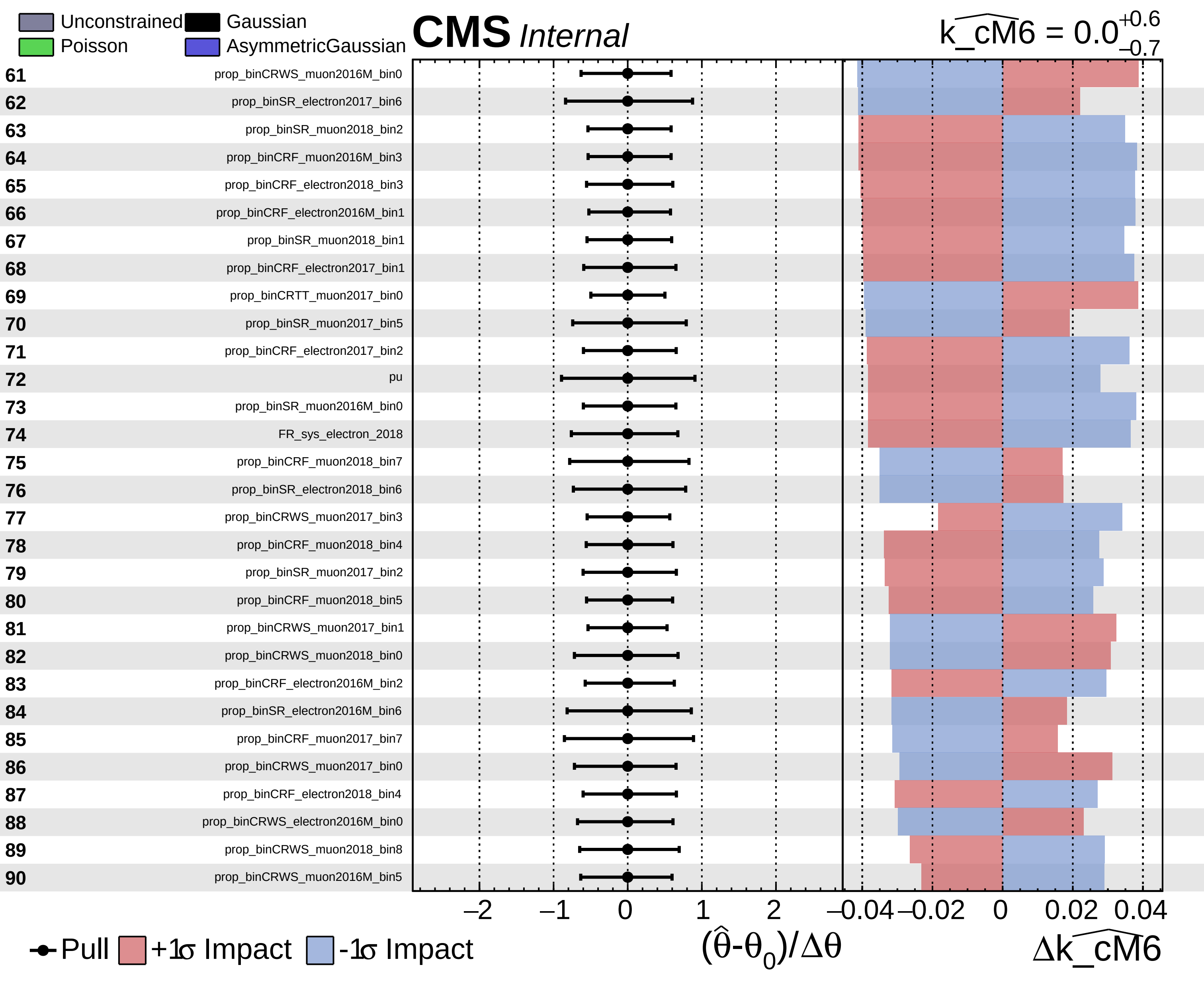


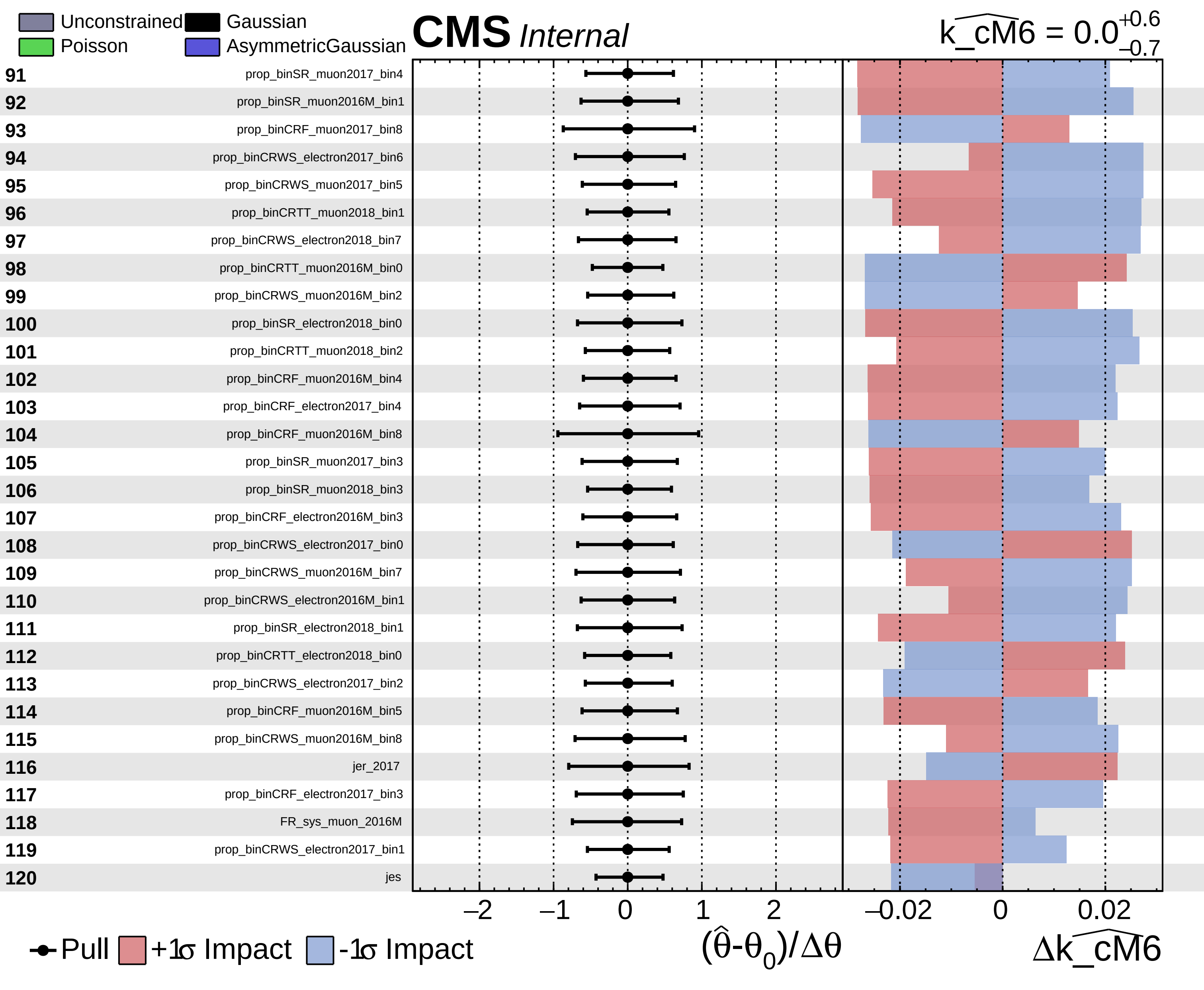
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM6 = 0.0 +0.6 −0.7
1 prop_binSR_muon2018_bin8
2 prop_binSR_muon2018_bin7
3 pdf_total
4 prop_binSR_muon2017_bin8
5 prop_binCRF_muon2018_bin0
6 prop_binCRF_muon2017_bin0
7 prop_binSR_electron2018_bin8
8 prop_binSR_muon2016M_bin8
9 prop_binSR_muon2017_bin7
10 prop_binSR_electron2018_bin7
11 FRest_muon_2017 1.00 +0.27 −0.20
12 TTest_electron_2016M 1.00 +0.24 −0.22
13 prop_binCRF_muon2016M_bin0
14 FRest_electron_2018 1.00 +0.25 −0.18
15 prop_binCRF_electron2018_bin0
16 TTest_muon_2016M 1.00 +0.23 −0.22
17 DYest_muon_2018 1.0 +0.5 −0.4
18 DYest_electron_2018 1.0 +0.6 −0.4
19 prop_binSR_muon2016M_bin7
20 TTest_electron_2017 1.00 +0.21 −0.21
21 DYest_muon_2016M 1.0 +0.6 −0.5
22 prop_binCRF_electron2017_bin0
23 DYest_muon_2017 1.0 +0.6 −0.5
24 prop_binCRF_muon2018_bin1
25 prop_binSR_electron2016M_bin8
26 prop_binCRF_muon2018_bin2
27 TTest_muon_2017 1.00 +0.20 −0.20
28 TTest_electron_2018 1.00 +0.18 −0.18
29 TTest_muon_2018 1.00 +0.18 −0.18
30 prop_binSR_electron2017_bin8
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM6

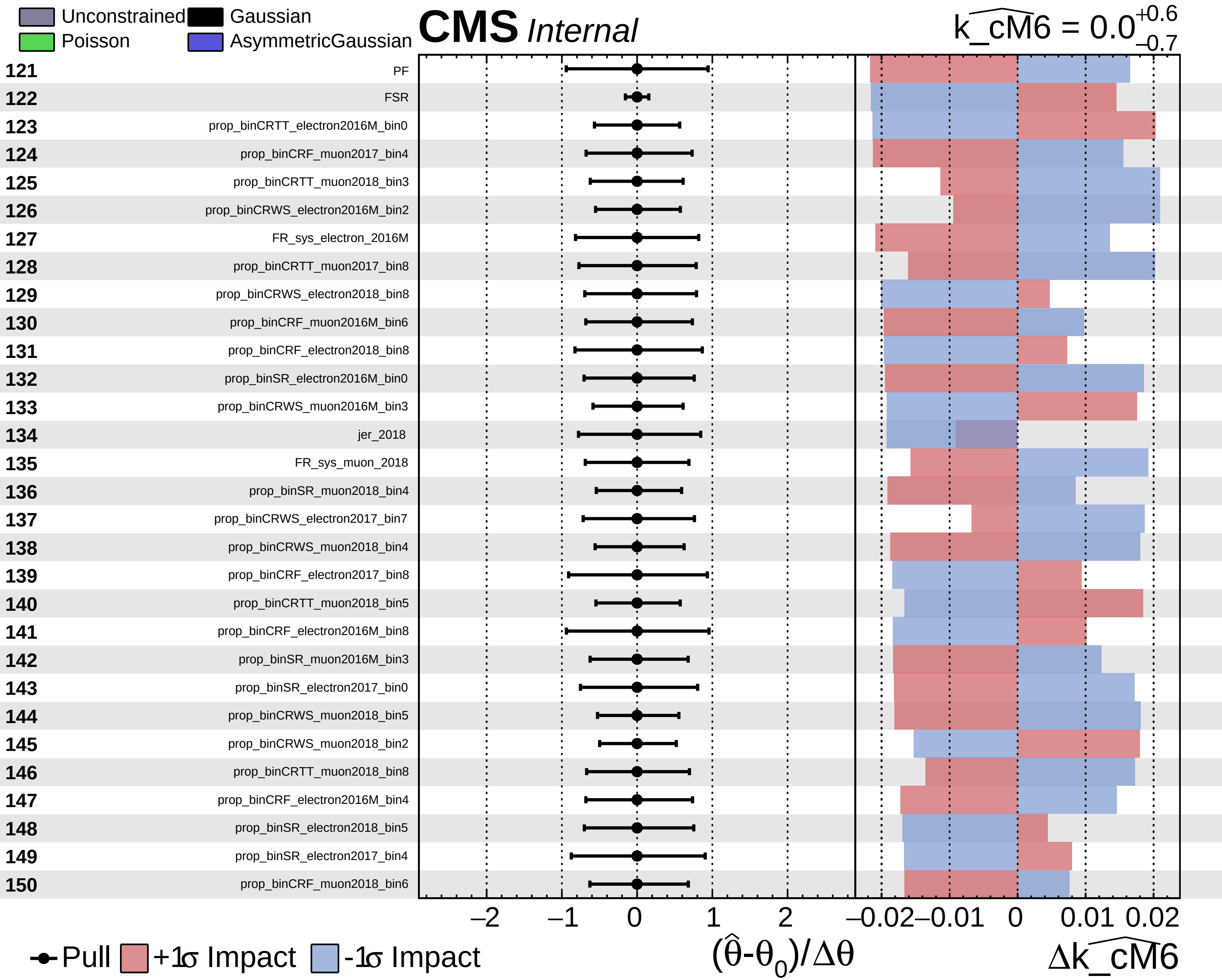


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM6 = 0.0 +0.6 −0.7
31 prop_binCRWS_muon2017_bin6
32 DYest_electron_2016M 1.0 +0.7 −0.7
33 FRest_muon_2018 1.00 +0.26 −0.21
34 prop_binSR_muon2018_bin0
35 prop_binCRF_electron2016M_bin0
36 FRest_muon_2016M 1.00 +0.25 −0.19
37 DYest_electron_2017 1.0 +0.8 −0.7
38 prop_binCRWS_electron2018_bin6
39 prop_binSR_electron2017_bin7
40 prop_binCRF_muon2017_bin1
41 prop_binCRF_muon2016M_bin1
42 prop_binSR_electron2016M_bin7
43 prop_binSR_muon2017_bin6
44 prop_binCRWS_muon2016M_bin4
45 prop_binSR_muon2016M_bin6
46 FRest_electron_2016M 1.00 +0.29 −0.21
47 prop_binCRF_muon2017_bin2
48 prop_binCRWS_muon2018_bin6
49 prop_binCRF_muon2018_bin3
50 prop_binSR_muon2017_bin0
51 prop_binSR_muon2018_bin6
52 prop_binCRF_electron2018_bin1
53 FRest_electron_2017 1.00 +0.28 −0.21
54 prop_binCRF_muon2018_bin8
55 prop_binCRTT_muon2018_bin0
56 prop_binCRF_electron2018_bin2
57 prop_binCRF_muon2016M_bin2
58 prop_binCRF_muon2017_bin3
59 FR_sys_muon_2017
60 prop_binSR_muon2017_bin1
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM6}$

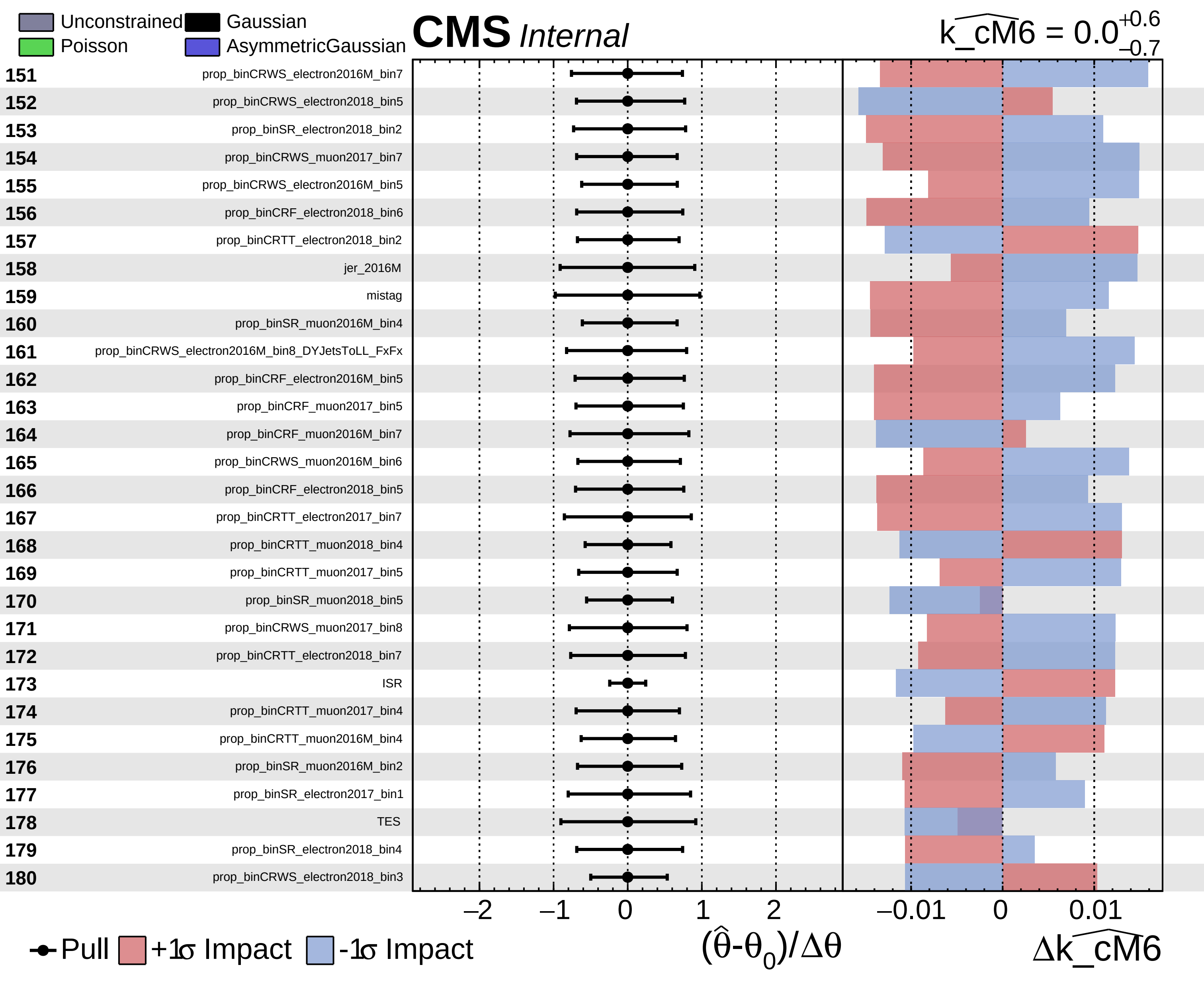


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM6} = 0.0^{+0.6}_{-0.7}$
61 prop_binCRWS_muon2016M_bin0
62 prop_binSR_electron2017_bin6
63 prop_binSR_muon2018_bin2
64 prop_binCRF_muon2016M_bin3
65 prop_binCRF_electron2018_bin3
66 prop_binCRF_electron2016M_bin1
67 prop_binSR_muon2018_bin1
68 prop_binCRF_electron2017_bin1
69 prop_binCRTT_muon2017_bin0
70 prop_binSR_muon2017_bin5
71 prop_binCRF_electron2017_bin2
72 pu
73 prop_binSR_muon2016M_bin0
74 FR_sys_electron_2018
75 prop_binCRF_muon2018_bin7
76 prop_binSR_electron2018_bin6
77 prop_binCRWS_muon2017_bin3
78 prop_binCRF_muon2018_bin4
79 prop_binSR_muon2017_bin2
80 prop_binCRF_muon2018_bin5
81 prop_binCRWS_muon2017_bin1
82 prop_binCRWS_muon2018_bin0
83 prop_binCRF_electron2016M_bin2
84 prop_binSR_electron2016M_bin6
85 prop_binCRF_muon2017_bin7
86 prop_binCRWS_muon2017_bin0
87 prop_binCRF_electron2018_bin4
88 prop_binCRWS_electron2016M_bin0
89 prop_binCRWS_muon2018_bin8
90 prop_binCRWS_muon2016M_bin5
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM6}$

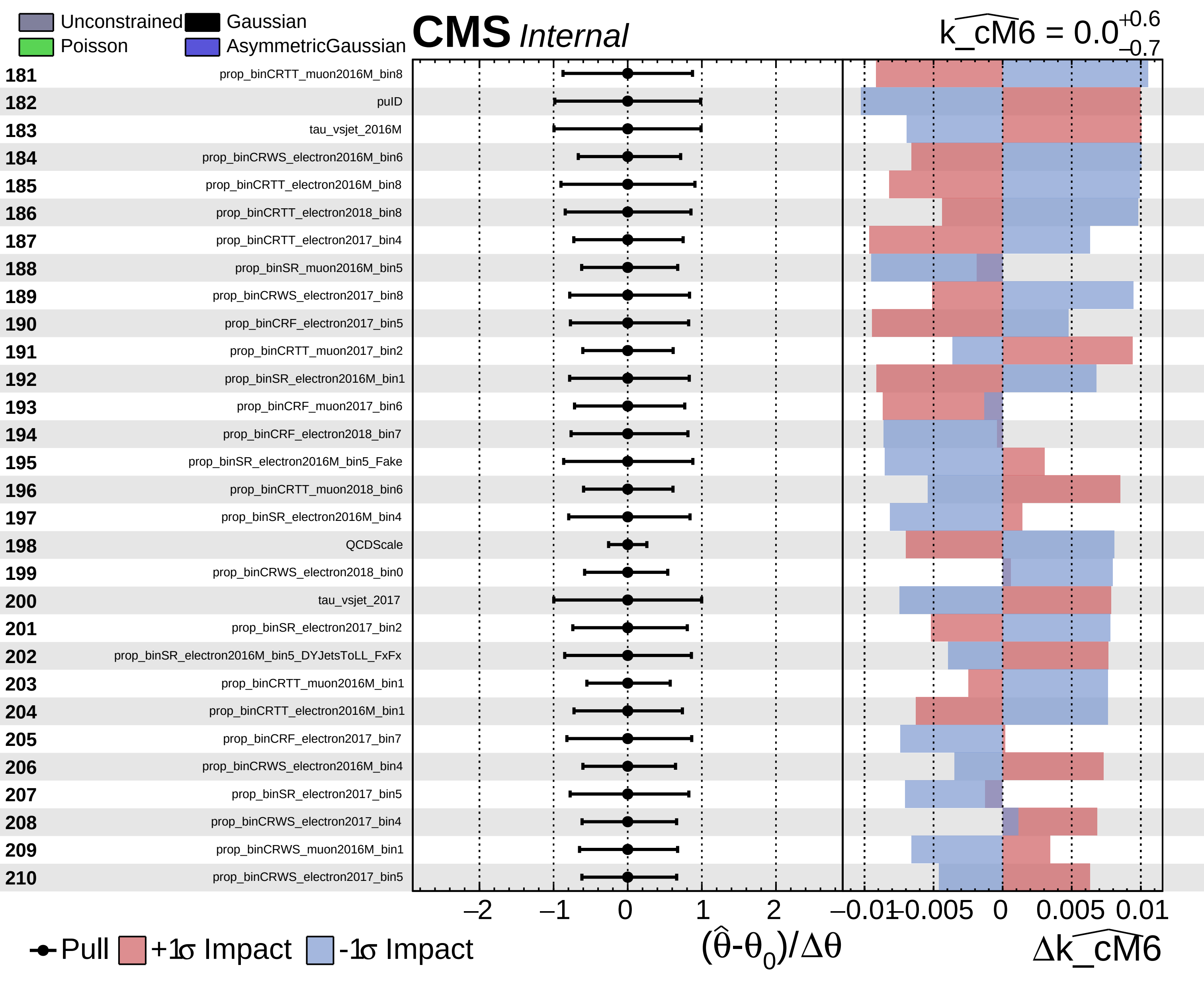

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM6} = 0.0^{+0.6}_{-0.7}$
91 prop_binSR_muon2017_bin4
92 prop_binSR_muon2016M_bin1
93 prop_binCRF_muon2017_bin8
94 prop_binCRWS_electron2017_bin6
95 prop_binCRWS_muon2017_bin5
96 prop_binCRTT_muon2018_bin1
97 prop_binCRWS_electron2018_bin7
98 prop_binCRTT_muon2016M_bin0
99 prop_binCRWS_muon2016M_bin2
100 prop_binSR_electron2018_bin0
101 prop_binCRTT_muon2018_bin2
102 prop_binCRF_muon2016M_bin4
103 prop_binCRF_electron2017_bin4
104 prop_binCRF_muon2016M_bin8
105 prop_binSR_muon2017_bin3
106 prop_binSR_muon2018_bin3
107 prop_binCRF_electron2016M_bin3
108 prop_binCRWS_electron2017_bin0
109 prop_binCRWS_muon2016M_bin7
110 prop_binCRWS_electron2016M_bin1
111 prop_binSR_electron2018_bin1
112 prop_binCRTT_electron2018_bin0
113 prop_binCRWS_electron2017_bin2
114 prop_binCRF_muon2016M_bin5
115 prop_binCRWS_muon2016M_bin8
116 jer_2017
117 prop_binCRF_electron2017_bin3
118 FR_sys_muon_2016M
119 prop_binCRWS_electron2017_bin1
120 jes
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM6}$

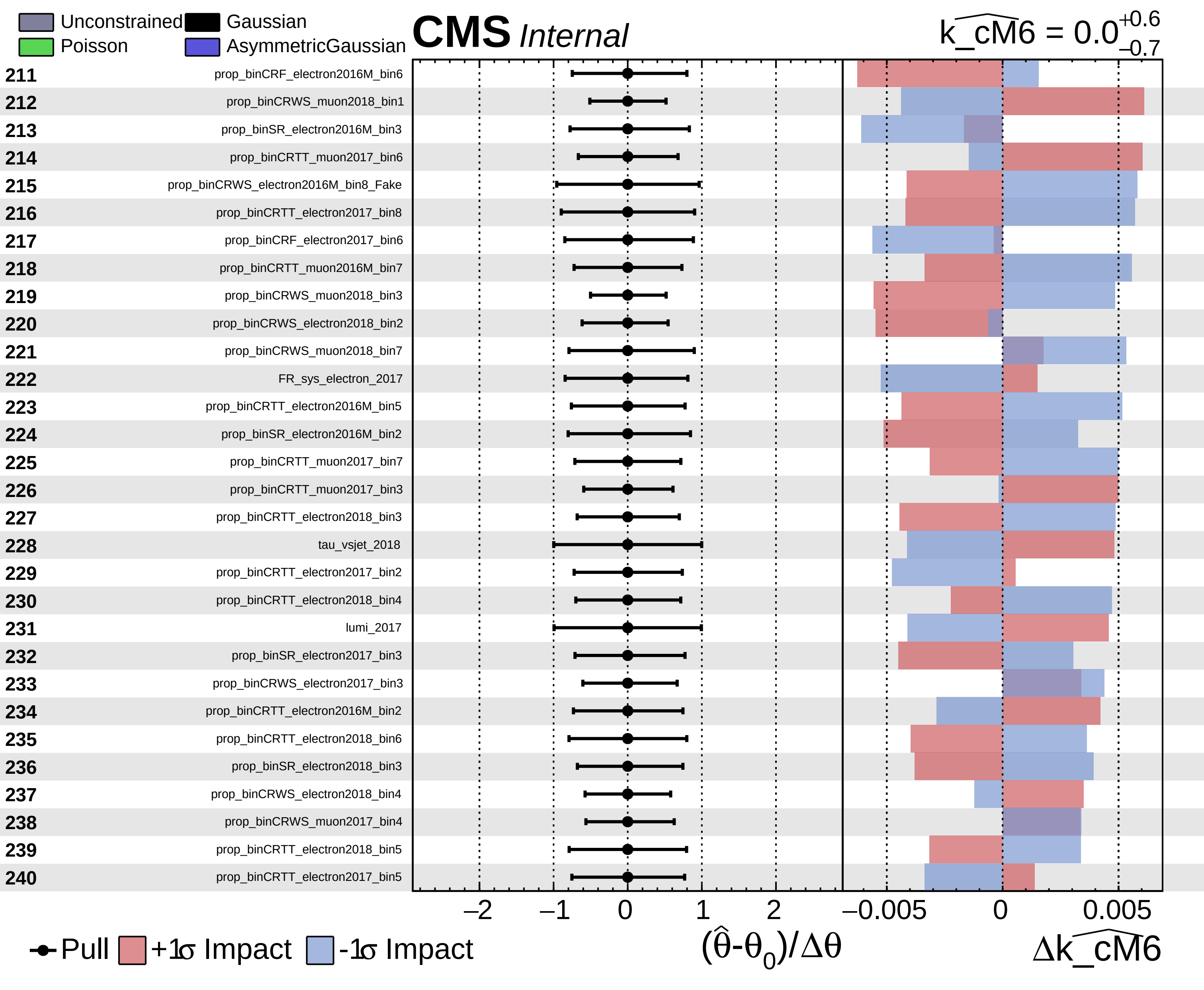


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM6} = 0.0^{+0.6}_{-0.7}$
121 PF
122 FSR
123 prop_binCRTT_electron2016M_bin0
124 prop_binCRF_muon2017_bin4
125 prop_binCRTT_muon2018_bin3
126 prop_binCRWS_electron2016M_bin2
127 FR_sys_electron_2016M
128 prop_binCRTT_muon2017_bin8
129 prop_binCRWS_electron2018_bin8
130 prop_binCRF_muon2016M_bin6
131 prop_binCRF_electron2018_bin8
132 prop_binSR_electron2016M_bin0
133 prop_binCRWS_muon2016M_bin3
134 jer_2018
135 FR_sys_muon_2018
136 prop_binSR_muon2018_bin4
137 prop_binCRWS_electron2017_bin7
138 prop_binCRWS_muon2018_bin4
139 prop_binCRF_electron2017_bin8
140 prop_binCRTT_muon2018_bin5
141 prop_binCRF_electron2016M_bin8
142 prop_binSR_muon2016M_bin3
143 prop_binSR_electron2017_bin0
144 prop_binCRWS_muon2018_bin5
145 prop_binCRWS_muon2018_bin2
146 prop_binCRTT_muon2018_bin8
147 prop_binCRF_electron2016M_bin4
148 prop_binSR_electron2018_bin5
149 prop_binSR_electron2017_bin4
150 prop_binCRF_muon2018_bin6
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM6}$

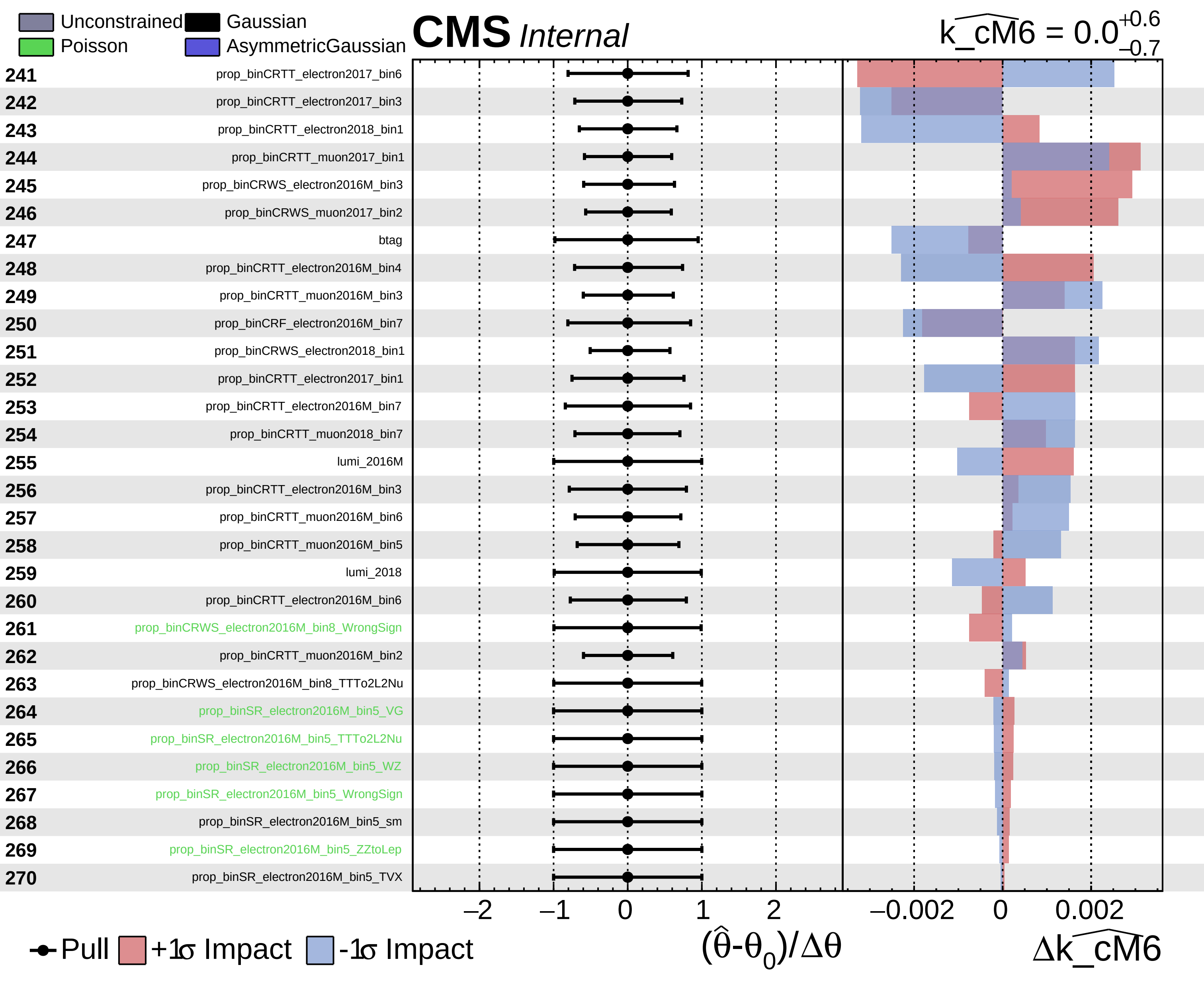


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM6 = 0.0 +0.6 −0.7
151 prop_binCRWS_electron2016M_bin7
152 prop_binCRWS_electron2018_bin5
153 prop_binSR_electron2018_bin2
154 prop_binCRWS_muon2017_bin7
155 prop_binCRWS_electron2016M_bin5
156 prop_binCRF_electron2018_bin6
157 prop_binCRTT_electron2018_bin2
158 jer_2016M
159 mistag
160 prop_binSR_muon2016M_bin4
161 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
162 prop_binCRF_electron2016M_bin5
163 prop_binCRF_muon2017_bin5
164 prop_binCRF_muon2016M_bin7
165 prop_binCRWS_muon2016M_bin6
166 prop_binCRF_electron2018_bin5
167 prop_binCRTT_electron2017_bin7
168 prop_binCRTT_muon2018_bin4
169 prop_binCRTT_muon2017_bin5
170 prop_binSR_muon2018_bin5
171 prop_binCRWS_muon2017_bin8
172 prop_binCRTT_electron2018_bin7
173 ISR
174 prop_binCRTT_muon2017_bin4
175 prop_binCRTT_muon2016M_bin4
176 prop_binSR_muon2016M_bin2
177 prop_binSR_electron2017_bin1
178 TES
179 prop_binSR_electron2018_bin4
180 prop_binCRWS_electron2018_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM6}$

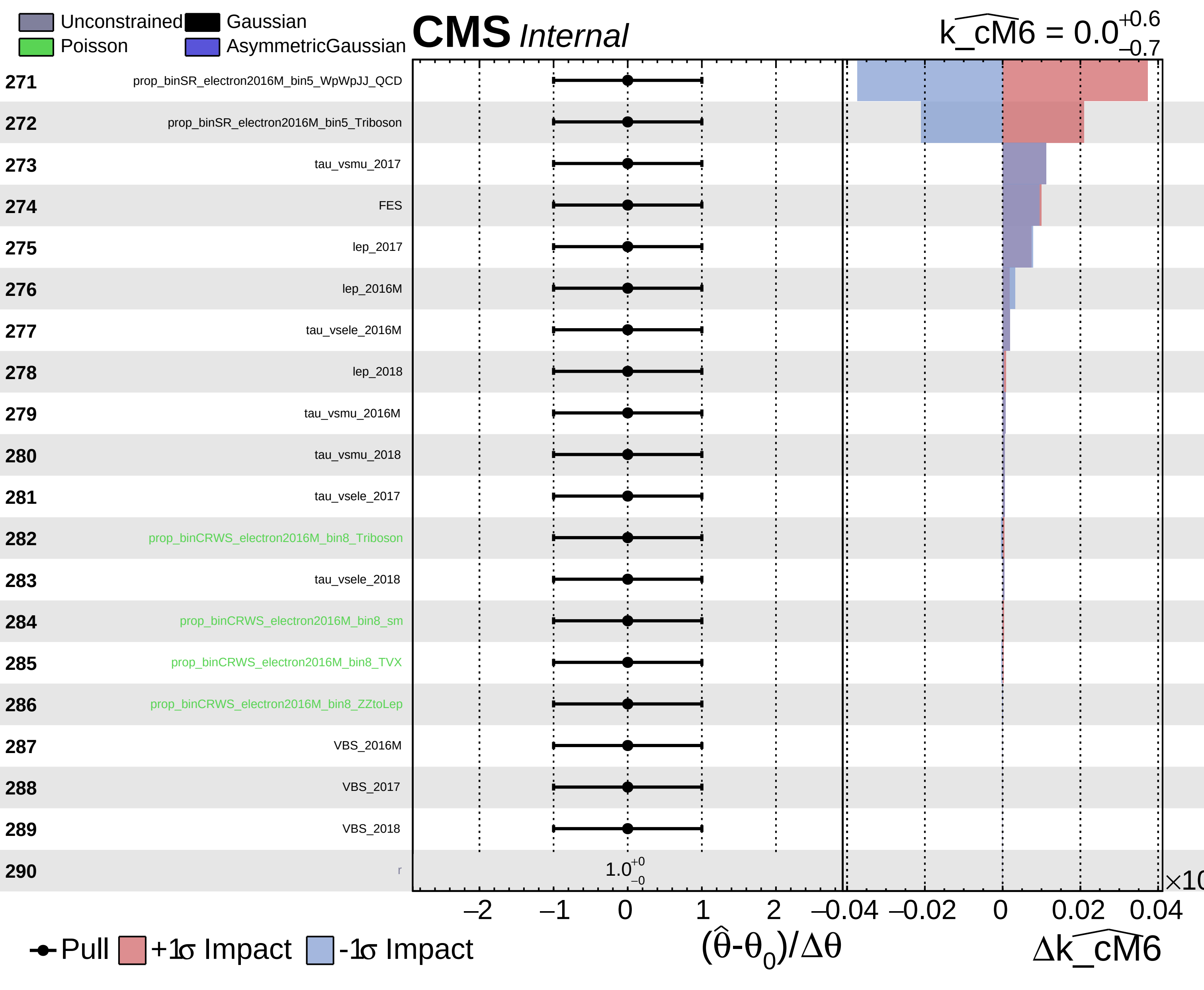

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM6 = 0.0 +0.6 −0.7
181 prop_binCRTT_muon2016M_bin8
182 puID
183 tau_vsjet_2016M
184 prop_binCRWS_electron2016M_bin6
185 prop_binCRTT_electron2016M_bin8
186 prop_binCRTT_electron2018_bin8
187 prop_binCRTT_electron2017_bin4
188 prop_binSR_muon2016M_bin5
189 prop_binCRWS_electron2017_bin8
190 prop_binCRF_electron2017_bin5
191 prop_binCRTT_muon2017_bin2
192 prop_binSR_electron2016M_bin1
193 prop_binCRF_muon2017_bin6
194 prop_binCRF_electron2018_bin7
195 prop_binSR_electron2016M_bin5_Fake
196 prop_binCRTT_muon2018_bin6
197 prop_binSR_electron2016M_bin4
198 QCDScale
199 prop_binCRWS_electron2018_bin0
200 tau_vsjet_2017
201 prop_binSR_electron2017_bin2
202 prop_binSR_electron2016M_bin5_DYJetsToLL_FxFx
203 prop_binCRTT_muon2016M_bin1
204 prop_binCRTT_electron2016M_bin1
205 prop_binCRF_electron2017_bin7
206 prop_binCRWS_electron2016M_bin4
207 prop_binSR_electron2017_bin5
208 prop_binCRWS_electron2017_bin4
209 prop_binCRWS_muon2016M_bin1
210 prop_binCRWS_electron2017_bin5
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM6



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM6 = 0.0 +0.6 −0.7
211 prop_binCRF_electron2016M_bin6
212 prop_binCRWS_muon2018_bin1
213 prop_binSR_electron2016M_bin3
214 prop_binCRTT_muon2017_bin6
215 prop_binCRWS_electron2016M_bin8_Fake
216 prop_binCRTT_electron2017_bin8
217 prop_binCRF_electron2017_bin6
218 prop_binCRTT_muon2016M_bin7
219 prop_binCRWS_muon2018_bin3
220 prop_binCRWS_electron2018_bin2
221 prop_binCRWS_muon2018_bin7
222 FR_sys_electron_2017
223 prop_binCRTT_electron2016M_bin5
224 prop_binSR_electron2016M_bin2
225 prop_binCRTT_muon2017_bin7
226 prop_binCRTT_muon2017_bin3
227 prop_binCRTT_electron2018_bin3
228 tau_vsjet_2018
229 prop_binCRTT_electron2017_bin2
230 prop_binCRTT_electron2018_bin4
231 lumi_2017
232 prop_binSR_electron2017_bin3
233 prop_binCRWS_electron2017_bin3
234 prop_binCRTT_electron2016M_bin2
235 prop_binCRTT_electron2018_bin6
236 prop_binSR_electron2018_bin3
237 prop_binCRWS_electron2018_bin4
238 prop_binCRWS_muon2017_bin4
239 prop_binCRTT_electron2018_bin5
240 prop_binCRTT_electron2017_bin5
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM6



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM6} = 0.0^{+0.6}_{-0.7}$
241 prop_binCRTT_electron2017_bin6
242 prop_binCRTT_electron2017_bin3
243 prop_binCRTT_electron2018_bin1
244 prop_binCRTT_muon2017_bin1
245 prop_binCRWS_electron2016M_bin3
246 prop_binCRWS_muon2017_bin2
247 btag
248 prop_binCRTT_electron2016M_bin4
249 prop_binCRTT_muon2016M_bin3
250 prop_binCRF_electron2016M_bin7
251 prop_binCRWS_electron2018_bin1
252 prop_binCRTT_electron2017_bin1
253 prop_binCRTT_electron2016M_bin7
254 prop_binCRTT_muon2018_bin7
255 lumi_2016M
256 prop_binCRTT_electron2016M_bin3
257 prop_binCRTT_muon2016M_bin6
258 prop_binCRTT_muon2016M_bin5
259 lumi_2018
260 prop_binCRTT_electron2016M_bin6
261 prop_binCRWS_electron2016M_bin8_WrongSign
262 prop_binCRTT_muon2016M_bin2
263 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
264 prop_binSR_electron2016M_bin5_VG
265 prop_binSR_electron2016M_bin5_TTTo2L2Nu
266 prop_binSR_electron2016M_bin5_WZ
267 prop_binSR_electron2016M_bin5_WrongSign
268 prop_binSR_electron2016M_bin5_sm
269 prop_binSR_electron2016M_bin5_ZZtoLep
270 prop_binSR_electron2016M_bin5_TVX
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM6}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM6 = 0.0 +0.6 −0.7
271 prop_binSR_electron2016M_bin5_WpWpJJ_QCD
272 prop_binSR_electron2016M_bin5_Triboson
273 tau_vsmu_2017
274 FES
275 lep_2017
276 lep_2016M
277 tau_vsele_2016M
278 lep_2018
279 tau_vsmu_2016M
280 tau_vsmu_2018
281 tau_vsele_2017
282 prop_binCRWS_electron2016M_bin8_Triboson
283 tau_vsele_2018
284 prop_binCRWS_electron2016M_bin8_sm
285 prop_binCRWS_electron2016M_bin8_TVX
286 prop_binCRWS_electron2016M_bin8_ZZtoLep
287 VBS_2016M
288 VBS_2017
289 VBS_2018
290 r
1.0 +0 −0
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM6